



Give list of all the differential and integration formule for class 11th, class 12th and JEE mains and advance, no bullshit, to the point

Here is a **concise, exam-focused list** of all the essential differentiation and integration formulas you need for Class 11, Class 12, JEE Main, and JEE Advanced. This covers algebraic, trigonometric, exponential, logarithmic, and inverse trigonometric functions, as well as the main rules and properties. If you want a PDF or Obsidian-friendly export, let me know.

Note: For any function of the form $f(ax + b)$, divide the result by the derivative of $ax + b$ (i.e., by a).

Differentiation Formulas

Algebraic Functions

- $\frac{d}{dx}(x^n) = nx^{n-1}$
- $\frac{d}{dx}(k) = 0$ (where k is constant)
- $\frac{d}{dx}(a^x) = a^x \ln a$
- $\frac{d}{dx}(e^x) = e^x$
- $\frac{d}{dx}(\ln x) = \frac{1}{x}$
- $\frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}$

Trigonometric Functions

- $\frac{d}{dx}(\sin x) = \cos x$
- $\frac{d}{dx}(\cos x) = -\sin x$
- $\frac{d}{dx}(\tan x) = \sec^2 x$
- $\frac{d}{dx}(\cot x) = -\csc^2 x$
- $\frac{d}{dx}(\sec x) = \sec x \tan x$
- $\frac{d}{dx}(\csc x) = -\csc x \cot x$

Inverse Trigonometric Functions

- $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$
- $\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$
- $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$
- $\frac{d}{dx}(\cot^{-1} x) = -\frac{1}{1+x^2}$
- $\frac{d}{dx}(\sec^{-1} x) = \frac{1}{|x|\sqrt{x^2-1}}$
- $\frac{d}{dx}(\csc^{-1} x) = -\frac{1}{|x|\sqrt{x^2-1}}$

Differentiation Rules

- **Sum/Difference:** $\frac{d}{dx}[f(x) \pm g(x)] = f'(x) \pm g'(x)$
- **Product:** $\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$
- **Quotient:** $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$
- **Chain Rule:** $\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$

Integration Formulas

Algebraic Functions

- $\int x^n dx = \frac{x^{n+1}}{n+1} + C, n \neq -1$
- $\int \frac{1}{x} dx = \ln|x| + C$
- $\int k dx = kx + C$
- $\int a^x dx = \frac{a^x}{\ln a} + C$
- $\int e^x dx = e^x + C$

Trigonometric Functions

- $\int \sin x dx = -\cos x + C$
- $\int \cos x dx = \sin x + C$
- $\int \sec^2 x dx = \tan x + C$
- $\int \csc^2 x dx = -\cot x + C$
- $\int \sec x \tan x dx = \sec x + C$
- $\int \csc x \cot x dx = -\csc x + C$
- $\int \tan x dx = -\ln|\cos x| + C$
- $\int \cot x dx = \ln|\sin x| + C$
- $\int \sec x dx = \ln|\sec x + \tan x| + C$

- $\int \csc x dx = \ln |\csc x - \cot x| + C$

Inverse Trigonometric Functions

- $\int \sin^{-1} x dx = x \sin^{-1} x + \sqrt{1 - x^2} + C$
- $\int \cos^{-1} x dx = x \cos^{-1} x - \sqrt{1 - x^2} + C$
- $\int \tan^{-1} x dx = x \tan^{-1} x - \frac{1}{2} \ln(1 + x^2) + C$
- $\int \cot^{-1} x dx = x \cot^{-1} x + \frac{1}{2} \ln(1 + x^2) + C$
- $\int \sec^{-1} x dx = x \sec^{-1} x - \ln |x + \sqrt{x^2 - 1}| + C$
- $\int \csc^{-1} x dx = x \csc^{-1} x + \ln |x + \sqrt{x^2 - 1}| + C$

Logarithmic and Exponential Functions

- $\int \ln x dx = x \ln x - x + C$
- $\int e^{ax} dx = \frac{1}{a} e^{ax} + C$
- $\int a^{ax} dx = \frac{1}{a \ln a} a^{ax} + C$

Standard Substitutions

- $\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + C$
- $\int \frac{1}{a^2 - x^2} dx = \frac{1}{2a} \ln \left| \frac{a+x}{a-x} \right| + C$
- $\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \left(\frac{x}{a} \right) + C$
- $\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln |x + \sqrt{x^2 + a^2}| + C$
- $\int \frac{1}{x^2 - a^2} dx = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right| + C$

Integration Rules

- **Sum/Difference:** $\int [f(x) \pm g(x)] dx = \int f(x) dx \pm \int g(x) dx$
- **Integration by Parts:** $\int u dv = u \int v dx - \int \left(\frac{du}{dx} \int v dx \right) dx$
- **Substitution:** If $x = g(t)$, then $\int f(x) dx = \int f(g(t)) g'(t) dt$
- **Partial Fractions:** Used for rational functions.

Definite Integrals

- $\int_a^b f(x) dx = F(b) - F(a)$, where $F(x)$ is any antiderivative of $f(x)$
- **Properties:**
 - $\int_a^a f(x) dx = 0$
 - $\int_a^b f(x) dx = - \int_b^a f(x) dx$
 - $\int_a^b f(x) dx = \int_a^b f(a + b - x) dx$

This covers all the core formulas for differentiation and integration as required for Class 11, 12, JEE Main, and Advanced.

If you want a more detailed breakdown (e.g., by topic or with examples), or a PDF/Obsidian export, just ask.

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1. <https://byjus.com/jee/important-integration-formulas-for-jee/>
2. [https://www.resonance.ac.in/sc/post/attachment/\(2233\)-gyan-sutra-m.pdf](https://www.resonance.ac.in/sc/post/attachment/(2233)-gyan-sutra-m.pdf)
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17. <https://allen.in/jee/maths/integral>